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PROJECT OVERVIEW

This project focused on the acquisition and computational analysis of biomedical data using MATLAB. Microscopy images of blood smears were processed to identify, segment, and quantify red blood cells (RBCs) and white blood cells (WBCs). The work also included microscope resolution characterization using a USAF 1951 target, comparison of imaging contrast methods, and biomedical signal processing including both time-domain and frequency-domain (Fourier transform) analysis of signals and microscopy images.

PROJECT OBJECTIVE

The objective of this project was to develop integrated image-processing and signal-processing workflows for quantitative biomedical analysis. The project combined microscopy, segmentation, and Fourier-based frequency analysis to extract meaningful biological and structural information from experimental datasets.

EXPERIMENTAL SETUP AND EQUIPMENT

The core instrumentation and toolsets integrated during this computational setup consisted of:

- Compound Light Microscope: Configured for strict Köhler Illumination alignment.
- Digital Camera System: CMOS image capturing element calibrated with a physical sensor pixel size of $3.6 \mu\text{m} \times 3.6 \mu\text{m}$ and an integrated $0.37\times$ optical de-magnifier before the sensor plane.
- USAF 1951 Resolution Target: Chrome-on-glass high-resolution standard spatial testing matrix.
- Biological Samples: Fixed blood smear samples stained via Wright's method alongside unstained human buccal epithelial (cheek) cells.
- MATLAB Software: Utilizing the Image Processing and Signal Processing Toolboxes.

METHODOLOGY

1. Resolution Characterization

Images of a USAF 1951 resolution target were captured using 4x, 10x, and 40x objectives. System resolution was evaluated by identifying the smallest resolvable line pairs. Theoretical resolution bounds were characterized using the standard classical Abbe resolution limit:

$$\delta = \lambda / (2NA)$$

Where:

- δ = resolution limit
- λ = wavelength of light
- NA = numerical aperture of the objective lens ($NA_{\{4x\}} = 0.10$, $NA_{\{10x\}} = 0.25$, $NA_{\{40x\}} = 0.65$)

During target acquisition, an exposure time of 2.519 ms was established for the 4x objective to maximize structural contrast boundaries across the element groups without pixel saturation.

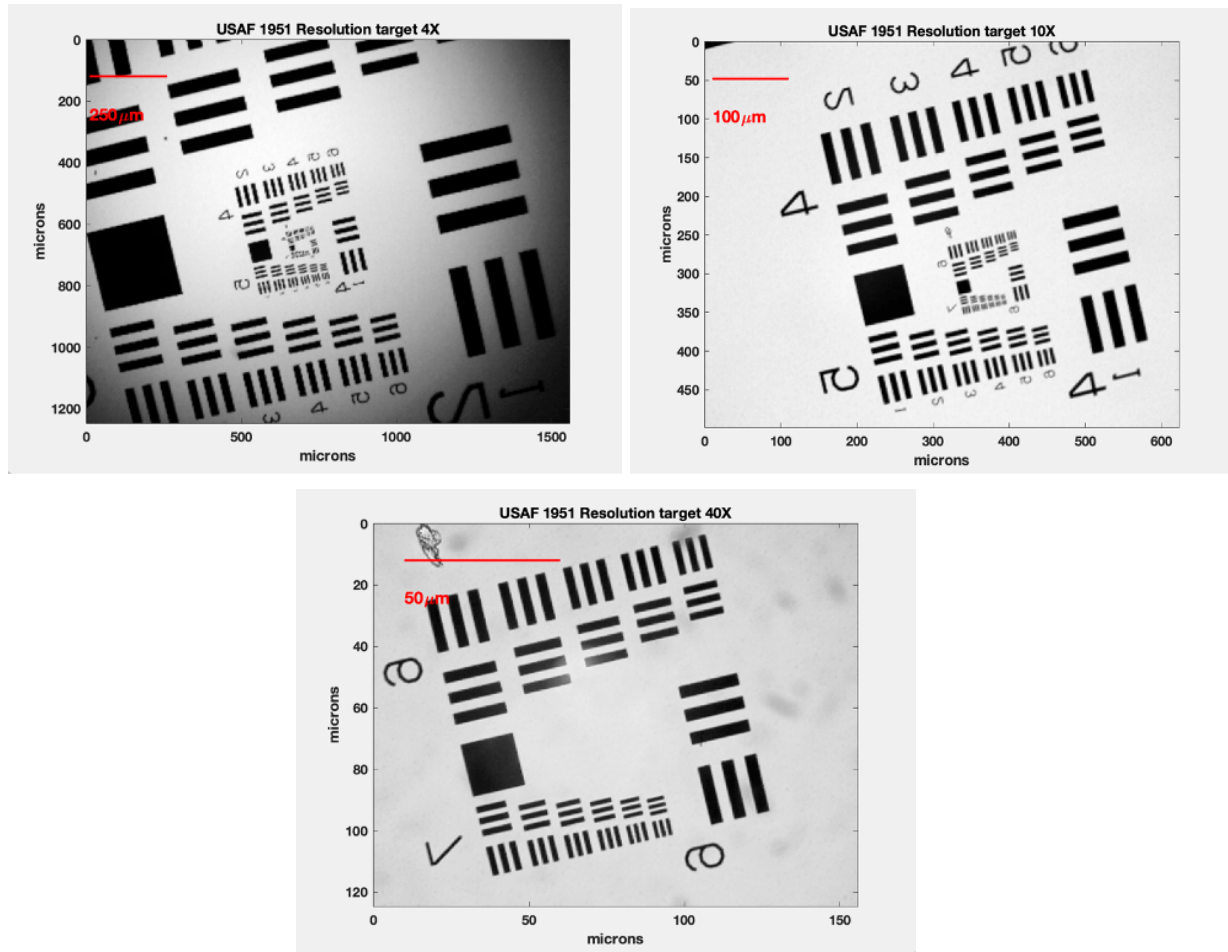


Figure 1: Captured element groups of the USAF 1951 calibration standard used to benchmark experimental spatial frequencies.

2. Contrast Mechanisms in Microscopy

Three distinct imaging modalities were structurally implemented and evaluated based on light scattering physics:

- **Brightfield Microscopy:** Standard transmission imaging used to visualize stained cellular structures. It provided excellent performance for Wright-stained smears but exhibited low amplitude contrast for completely unstained transparent cheek cells.
- **Phase Contrast Microscopy:** Converted subtle phase differences ($\Delta n \cdot t$) into visible amplitude/intensity variations. This significantly enhanced transparent biological structures, adding 3D-shadowed structural definitions to borders without cytotoxic staining.
- **Darkfield Microscopy:** Collected only scattered light by blocking the direct illumination beam with a central patch stop. This enhanced the visibility of small boundaries, organelles, and cell edges, which appeared brightly lit against a dark background.

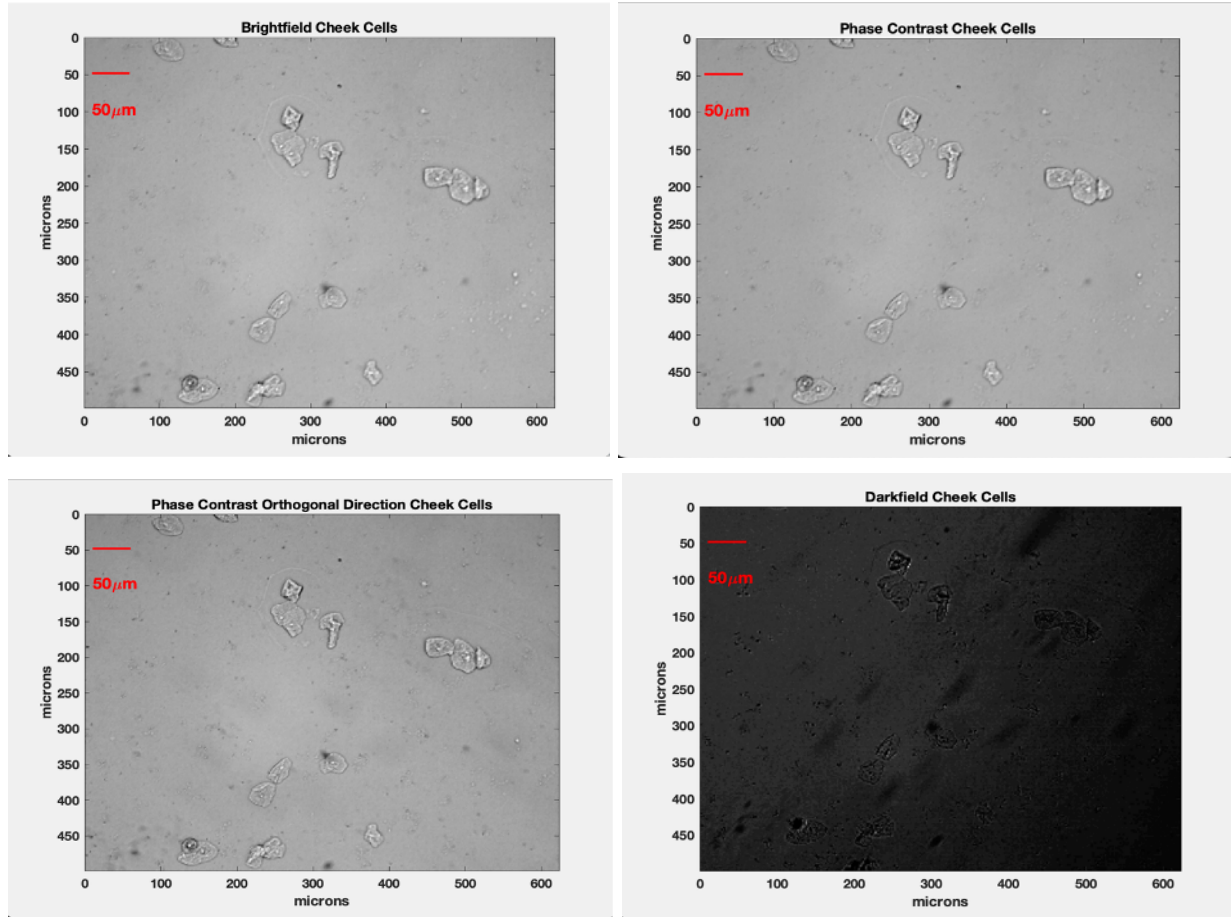


Figure 2: Morphological contrast variations of human buccal epithelial cells visualized across alternative optical path mechanisms.

In a brightfield image, small features or edges in the cells appear slightly darker than the surrounding bright background. Features appear slightly darker because they scatter some of the illumination light into angles that are so large that they are no longer collected by the microscope objective. Sample features lead to a reduction in light intensity compared to the bright background called brightfield microscopy. In an oblique illumination image, we are forcing light to be incident on the sample from an angle. It converts phase variations in light into intensity variations. Oblique illumination leads to a phase contrast which can reveal subtle structures in cells that are otherwise invisible. The cheek cells take on a 3D appearance because of shadowing effects.

The two oblique illumination images differ in the direction of the shadows and contrast effects due to the change in the angle of illumination. The first image has the shadow and bright regions along a specific axis, making certain features of the cheek cells stand more clearly. The second image will have shadows appearing in a different direction, rotated about 90 degrees

relative to the first image. This happens because blocking light from another direction alters how light interacts with cell structures.

In a darkfield image, the only light that can be seen is scattered light from the sample, since unscattered light misses the objective completely. The scattering must deviate the illumination light by a relatively large angle, meaning it can only arise from very small structures in the sample and nothing else. In a bright field image, light passes through the sample and reaches the objective directly, where the sample will appear mostly transparent. In a darkfield, features that were barely visible in brightfield become contrasted showing features a brightfield microscopy cannot capture. Features that are enhanced in a darkfield are tiny particles such as bacteria and edges of cells and internal structures more visible because of more scattering.

3. Blood Cell Segmentation and Quantification

Microscopy images of blood smears were processed in MATLAB using an automated tracking script. To streamline calculation across high-throughput datasets, a for loop was implemented to import sequentially numbered matrices (n = 1 to 9) matching the file format string:

```
filename = ['BloodSmear10x_' num2str(n) '.bmp'];
```

MATLAB AUTOMATED BINARIZATION AND PROCESSING PIPELINE

Step A: RGB Channel Separation — Parse raw blood smear frames into individual sub-color arrays.



Step B: Channel Optimization — Extract Green channel for RBC visibility; extract Red channel for WBC nucleus absorption.



Step C: Image Thresholding & Masking — Apply global intensity cutoff bounds to create binary masks.



Step D: Morphological Filtering — Fill internal lumens and apply area filters to eliminate noise artifacts.

The effective pixel area matrix was calibrated back to true physical metrics by defining the camera and optical paths. Given a camera pixel size of $3.6\text{ }\mu\text{m}$ and an added $0.37\times$ de-magnifier, the effective pixel sizing at a $10\times$ scanning magnification was modeled as:

$$\text{Effective Pixel Dimension} = 3.6\text{ }\mu\text{m} / (10 \times 0.37) = 0.97297\text{ }\mu\text{m} / \text{pixel}$$

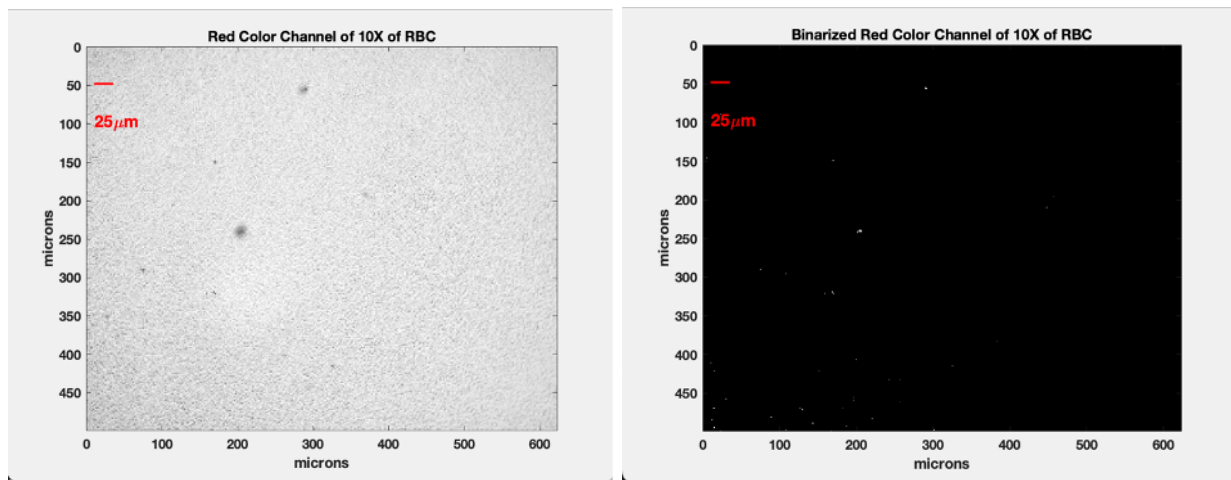


Figure 4: Red color channel of RBCs to find WBCs with a scale bar of $25\text{ }\mu\text{m}$ at $10\times$ magnification. The dark specs are not white blood cells and will appear as white blood cells in the binarized image. Binarized Red color channel to locate the points where WBCs sit with a scale bar of $25\text{ }\mu\text{m}$ at a magnification of $10\times$. The two largest white spots correspond to the shadows shown in the Red color channel, and are not white blood cells.

Number of WBCs in 10X image: _____7

Average Number of Pixels per WBC in 10X image #1: _____8

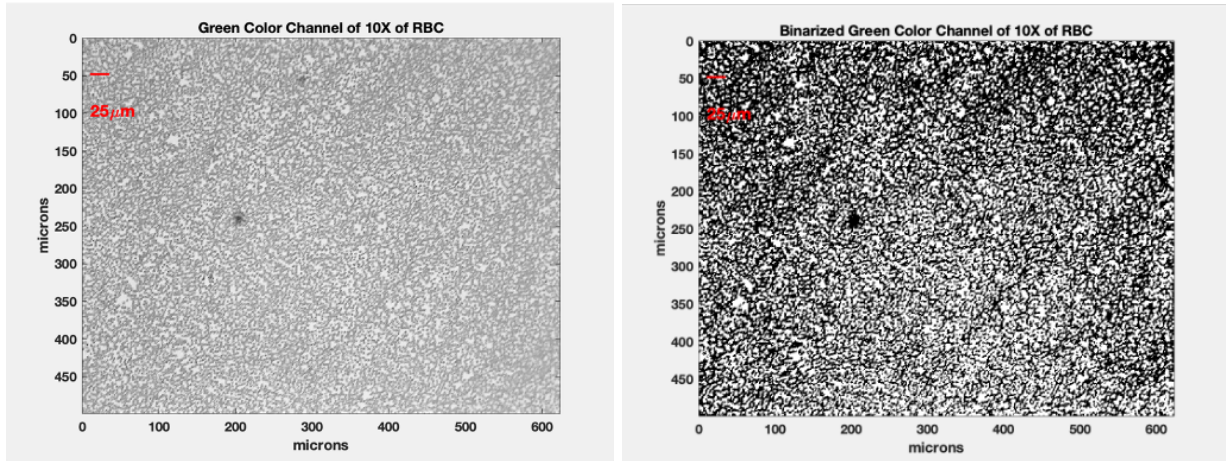


Figure 5: Green color channel(left) of RBCs and its binarized image(right) with a scale bar of 25 μm at 10X magnification.

Number of Pixels containing RBCs in 10X image #1: 170900

Number of RBCs in 10X image #1: 14242

$$\text{Number of Cells} = \frac{\text{Total Cells type Pixels}}{\text{Average Number of Pixels per cell Type}}$$

$$\text{Number of RBCs} = \frac{170,900}{12} = 14,242 \text{ RBCs}$$

$$\text{Number of WBCs} = \frac{58}{7} \sim 7 \text{ WBCs}$$

$$\frac{\text{WBCs}}{\text{mm}^2} = \frac{7 \text{ WBCs}}{.307 \text{ mm}^2} = 27 \frac{\text{WBCs}}{\text{mm}^2}$$

$$\frac{\text{RBCs}}{\text{mm}^2} = \frac{14,242 \text{ RBCs}}{.307 \text{ mm}^2} = 46,390 \frac{\text{RBCs}}{\text{mm}^2}$$

5. IMAGE-BASED FOURIER TRANSFORM ANALYSIS (2D FFT)

Blood smear images were transformed into the frequency domain using a 2D Fast Fourier Transform (fft2 and fftshift) in MATLAB. To evaluate a real-world spatial distribution, the 40x RBC image was converted to an NTSC grayscale format, cropped into a perfectly square pixel configuration with an odd pixel length in both directions, and centered by subtracting its global mean to eliminate high-power DC baseline bias.

Log-scaled magnitude spectra were implemented for structural visualization. Biological interpretations revealed that the spatial distribution of red blood cells contributed to repetitive mid-frequency structural features, white blood cells appeared as localized high-intensity frequency disturbances, and random staining artifacts mapped primarily as low-frequency background noise components.

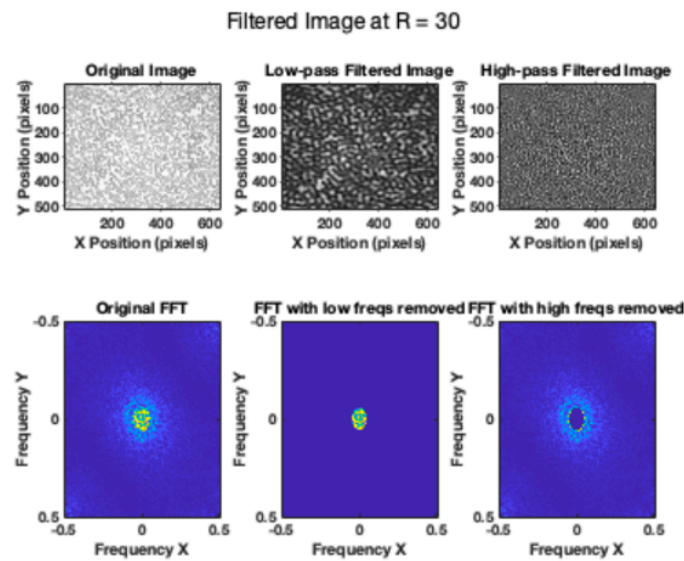


Figure 6: Filtering the 40X red blood cell image at radius = 30 pixels. The first column is the original image and the FFT of it. The second column is the Low pass filtered image with its corresponding FFT. The last column is the High pass filtered image with its corresponding FFT.

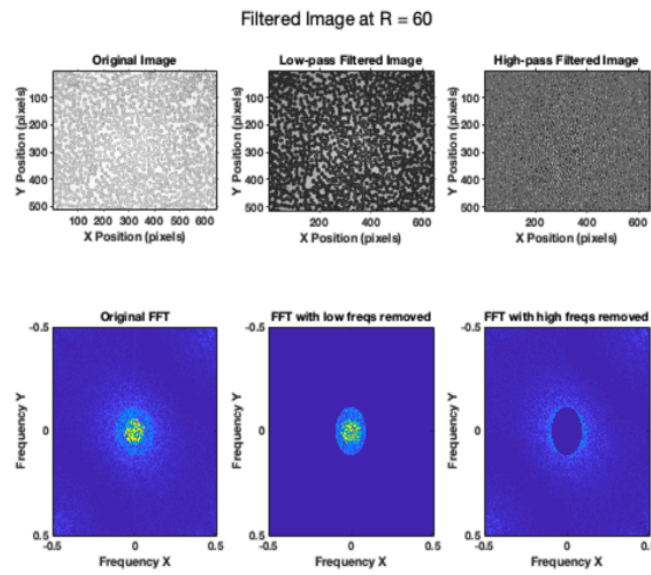


Figure 7: Filtering the 40X red blood cell image at radius = 60 pixels. The first column is the original image and the FFT of it. The second column is the Low pass filtered image with its corresponding FFT. The last column is the High pass filtered image with its corresponding FFT.

High spatial frequencies show fine details, small objects, sharp edges, and noise. These can correspond to the red blood cells with their outlines and even small organelles. Low spatial frequencies represent larger shapes, the overall structure such as a similar color or intensity. The background of the red blood cells can be represented by low spatial frequencies.

The strategy began with converting the image to grayscale and subtracting its mean to the frequency components of the original image. An FFT was composed to get the frequency domains. In low pass filters, a mask was created to keep low frequencies near the center or inside the radius, zeroing high frequencies. The high pass filter zeroes the center keeping high frequencies on the outside. The mask was applied to the FFT, shifted, and then inversely transformed back to the spatial domain.

Increasing the radius allows for more frequencies to pass through the filter. In a low pass, a larger radius reduces blurring keeping more detail in the image. In a high pass, a larger radius makes the image lose structure and look rougher. With a smaller radius, low pass filters would have more blurring because it keeps less frequencies. A smaller radius in high pass filters means more frequencies are removed. Images will show more sharper edges, but look more noisy.

REFLECTION

This project demonstrated how computational methods can be applied across both imaging and signal domains in biomedical engineering. Microscopy-based segmentation enabled quantitative biological measurement, while Fourier analysis in 2D domains provided deeper insight into signal behavior and spatial structure. Together, these methods highlight the importance of frequency-domain analysis in modern biomedical data processing.

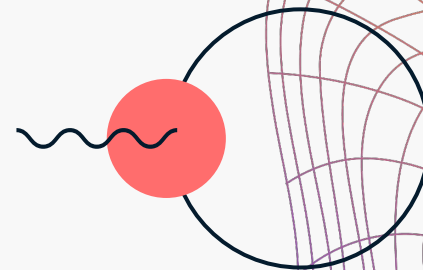


BE493

Section B1, Monday 2:30 - 5:15 pm

SpectraMax i3x Plate Reader

Christian Abou-Ezzi, Chizitere Chimezie, Kelly
Falcon, Parthi Mashru, Dylann Palmer

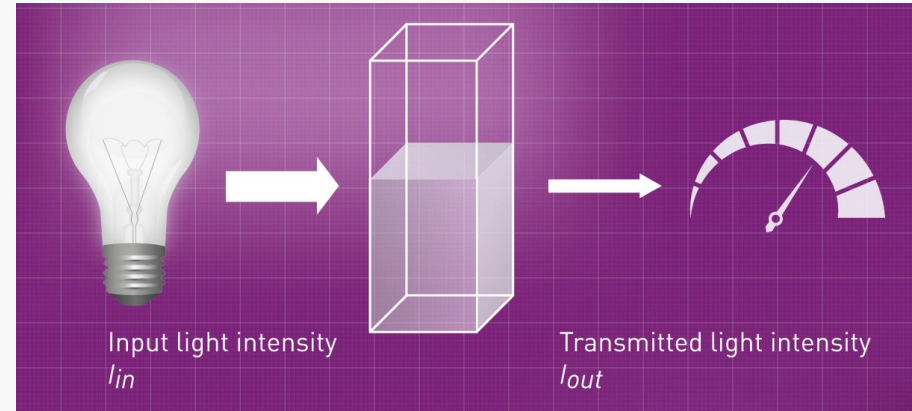


Absorbance

- Quantity of light absorbed by matter
- Measure of optical density
- Beer-Lambert Law
 - Absorbance vs Concentration
- Transmittance
 - Ratio of Transmitted Light to Incident Light

$$A = \epsilon bc = -\log(T)$$

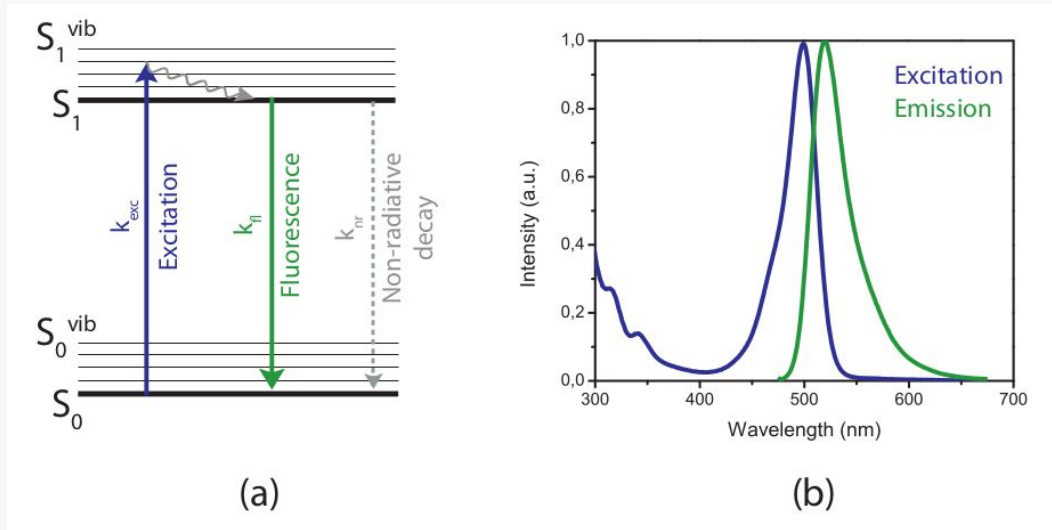
$$T = I_{\text{out}} / I_{\text{in}}$$



BMG Labtech 2025

Fluorescence

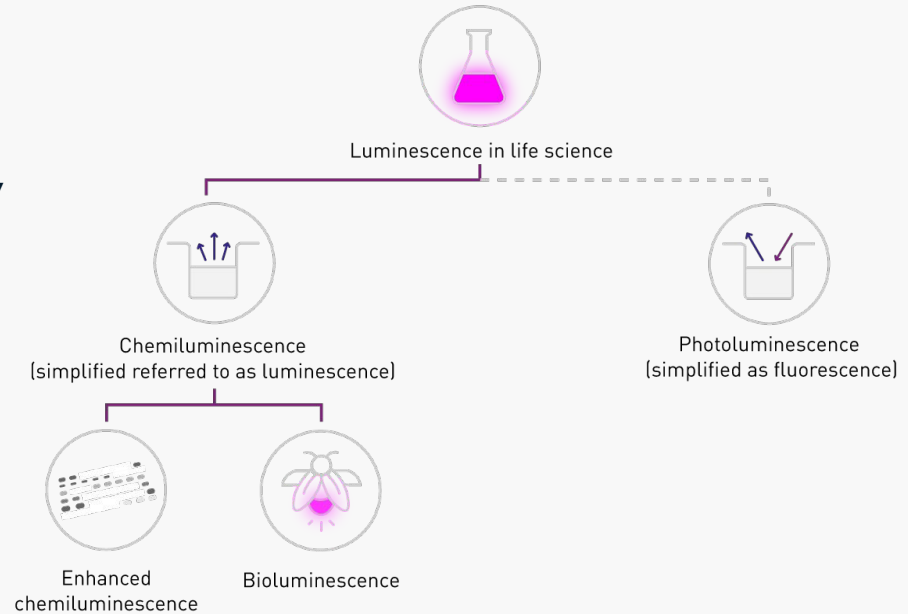
- Emission of light by matter after excitation
 - $\lambda_{\text{excitation}} < \lambda_{\text{emission}}$
- Stokes shift
- Measured in RFU



Scientific Volume Imaging 2025

Luminescence

- Emission of light from non thermal processes
 - Excitation not necessary
 - Chemiluminescence, Bioluminescence, Photoluminescence
- Measured in RLU



Major Components- External



Touchscreen – adjust plate holder
→

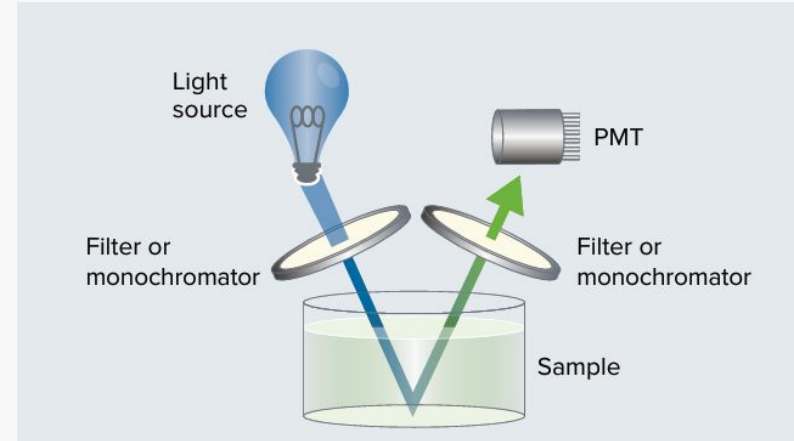


← Computer w/
Integrated Soft Pro
Max Software–
analyzes data
collected & helps
user control
parameters

Paracor Medical 2024

Major Components- Internal

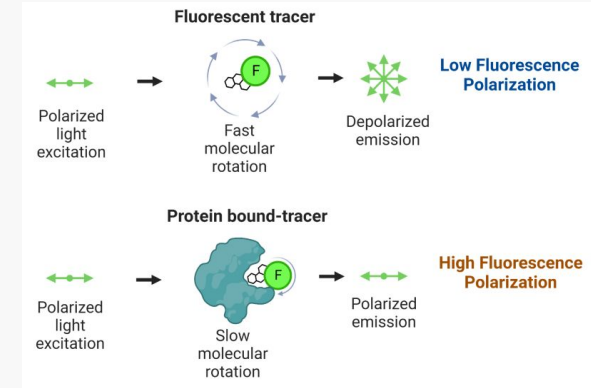
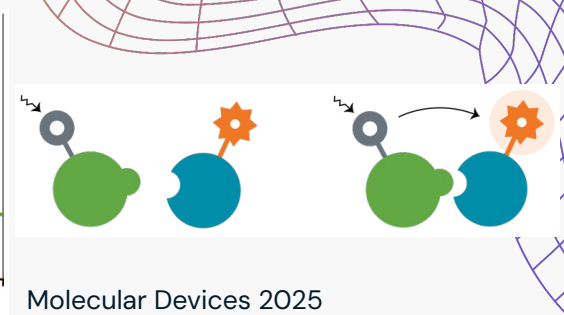
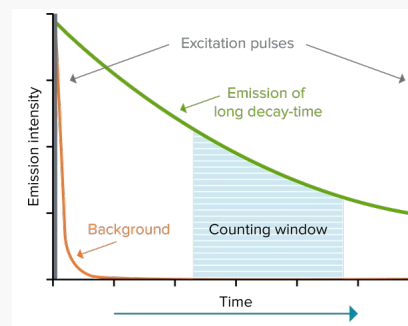
- **Light source= xenon flash lamp & LED**→ provides excitation light for absorbance & fluorescence
- **Optical filter/monochromator**→ selects specific wavelengths for emission & excitation
- **Photomultiplier tube (PMT)** → detects light from samples and converts to electrical signaling
- **Plate Holder**→ holds multiwell plate that holds specific amount of sample being measured



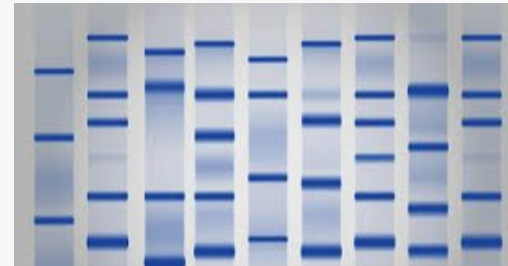
Molecular Devices 2021

Detection Modes

- Absorbance
- Fluorescence
- Luminescence
- Time-Resolved Fluorescence/
Time-Resolved Fluorescence
Resonance Energy Transfer
(TRF / TR-FRET)
- Fluorescence Polarization (FP)
- Western Blot



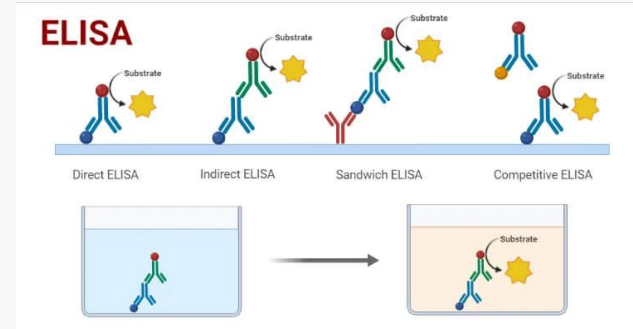
BPS Bioscience 2025



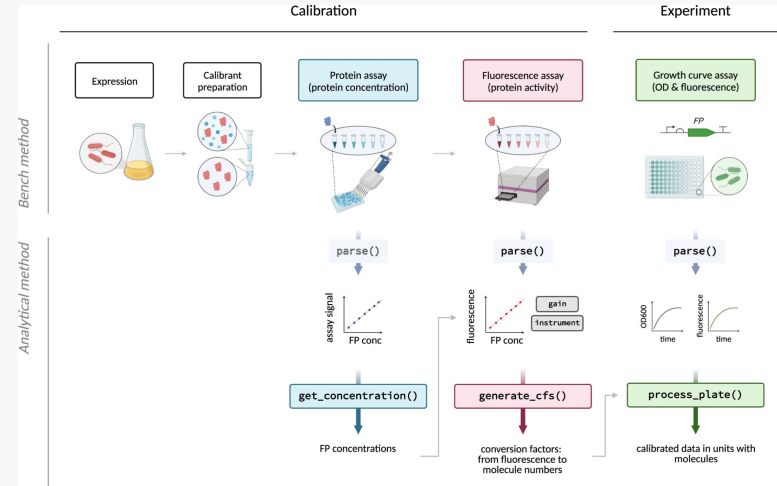
Boston Bio 2025

General Applications

- Cell Viability
- Cell Signaling
- Cell Migration
- Enzyme-Linked Immunosorbent Assay (ELISA)
- Microbiology and contaminant monitoring
- Nucleic acid (DNA/RNA) quantitation & analysis
- Protein quantitation & analysis



Microbe Notes 2022



Nature 2022

Applications in Research

1) Cold Atmospheric Plasma treatment effects on oxidative stress levels in sweet basil leaves.

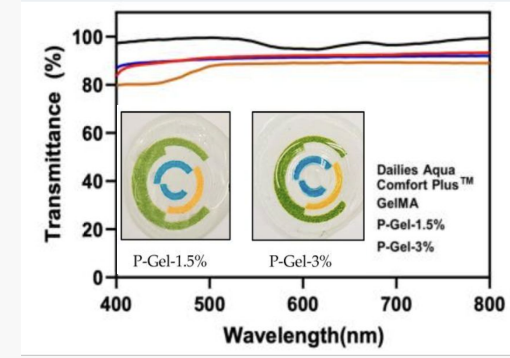
- ❑ Measure fluorescence to quantify Reactive Oxygen Species levels.
- ❑ Significant loss in fluorescence intensity → elevated ROS → CAP exposure increases oxidative stress.

2) 3D-printed contact lenses with polyvinyl alcohol release to treat dry eyes.

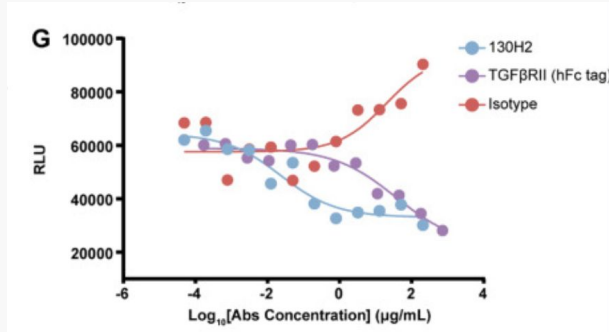
- ❑ UV-Vis absorbance spectroscopy for %T of lenses
- ❑ Reported transparency profiles of their hydrogels → Determined best lens design while also delivering PVA.

3) Blocking $\alpha\beta 8$ -mediated TGF- β signaling to reverse immunosuppression

- ❑ Measured decrease in RLU (Relative Luminescence Units)
- ❑ Antibody demonstrated strong TGF- β blocking activity → High potency in inhibiting TGF- β signaling.



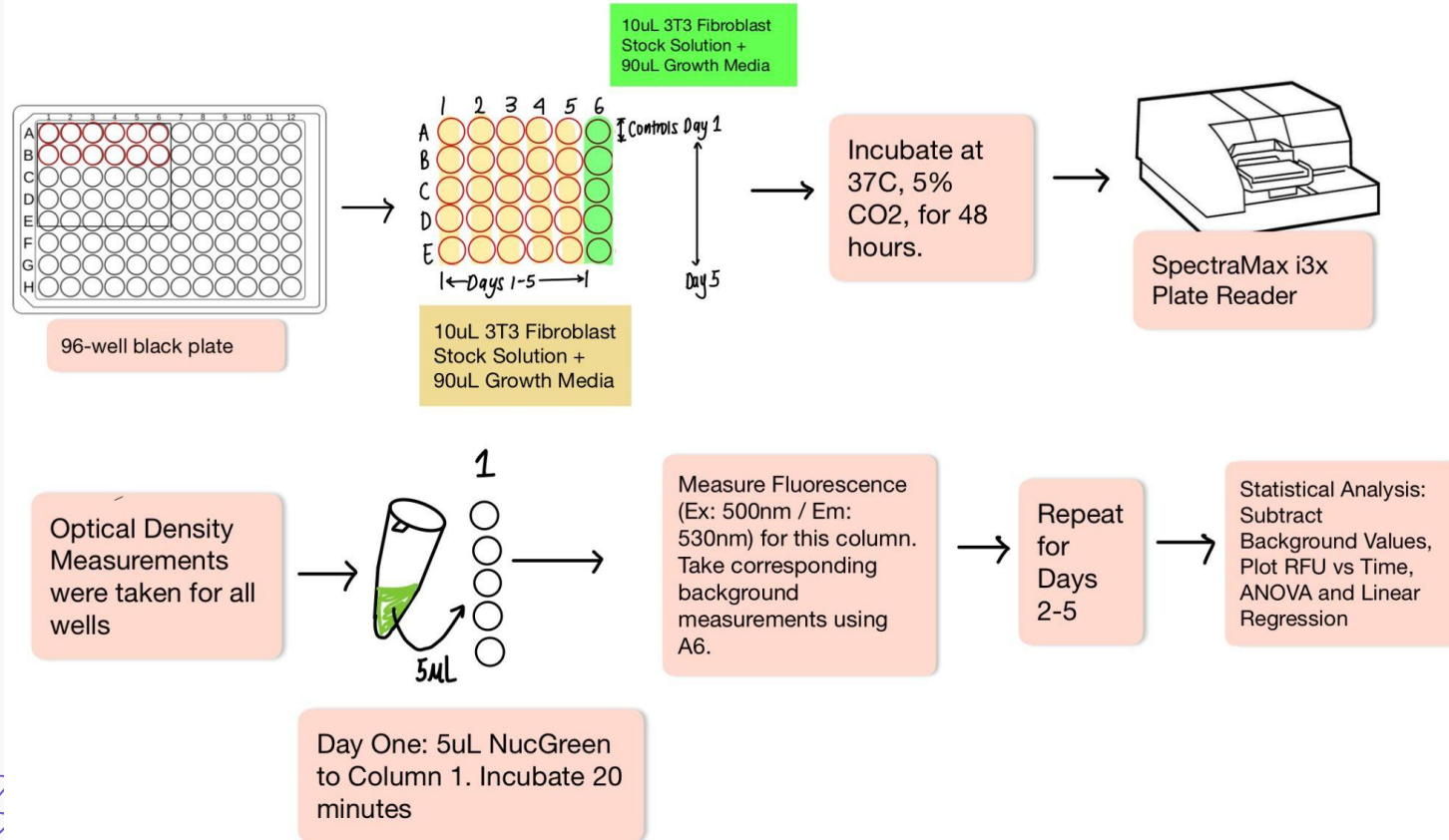
Piyush et al, Pharmaceuticals, 2025



CuiCui et al, Journal of Cancer Research, 2025



Experiment: Analysing cell death using Fluorescence in 3T3 Fibroblast cells



Optical Density-Absorbance



Table 1: Light Intensity reading from the plate reader

	1	2	3	4	5	6
A	0.116	0.118	0.133	0.12	0.109	0.106
B	0.125	0.16	0.155	0.159	0.155	0.147
C	0.132	0.176	0.15	0.157	0.151	0.136
D	0.121	0.143	0.153	0.156	0.15	0.142
E	0.115	0.111	0.128	0.121	0.124	0.112

- Light intensity that passed through the wells
- **Higher value** = more light passed through $\Rightarrow$ **less absorbance**
- **Lower value** = less light passed through $\Rightarrow$ **more absorbance**

Table 2: Percent Optical Density compared to the background average Intensity

	1	2	3	4	5	6
A	4.48%	3.74%	-1.46%	3.01%	7.18%	8.39%
B	1.23%	-9.49%	-8.11%	-9.22%	-8.11%	-5.81%
C	-1.13%	-13.63%	-6.69%	-8.67%	-6.97%	-2.43%
D	2.65%	-4.61%	-7.55%	-8.39%	-6.69%	-4.30%
E	4.85%	6.39%	0.20%	2.65%	1.58%	6.00%

- Intensity of light (I_1)
- average cell media background intensity (I_0)
- $\%OD = -\log(I_1/I_0) * 100$
- Negative $\Rightarrow$ **more light passed** $\Rightarrow$ less cells
- Positive $\Rightarrow$ **less light passed** $\Rightarrow$ more cells

Daily Measurements

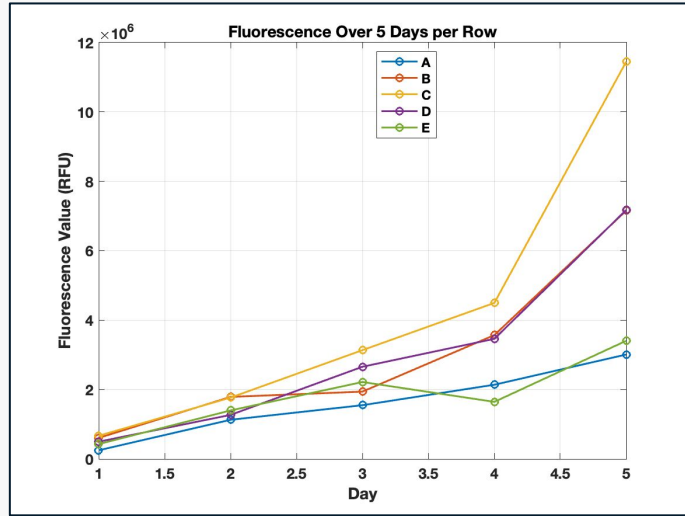


Figure 1: Fluorescence intensity over five days for each sample (A–E), plotted after background correction. Each line represents an individual sample's fluorescence trend across time.

- Anova P-Value 0.0002, statistically significant
- Shows growing signal intensity because of cell death
- complementing the average trends

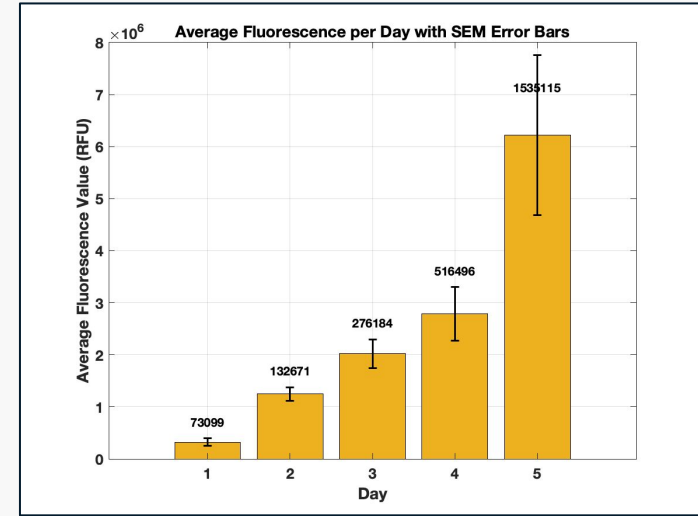


Figure 2: Average fluorescence intensity values per day across five samples after background correction. Bars represent the mean fluorescence values, and error bars indicate the standard error of the mean for each day.

- Reflecting variability among samples
- Error bars $\Rightarrow$ indicate standard error of the mean each day
- Upward trend in fluorescence
- Highest signal on day 5, most cell death

Regression Graphs

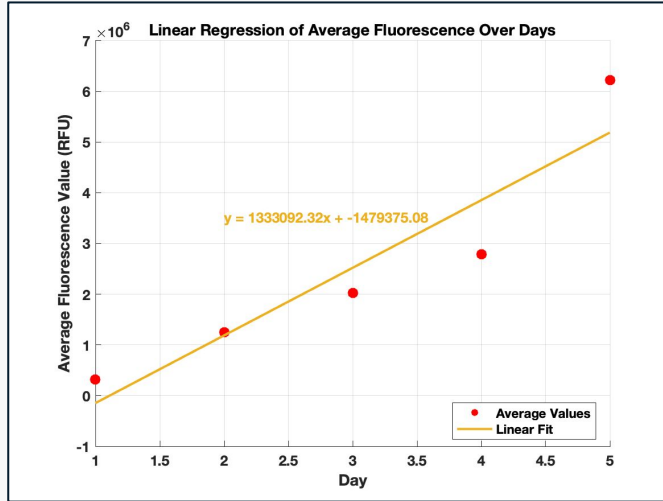


Figure 3: Linear regression of average fluorescence values over five days. A clear positive linear trend is observed, with a statistically significant slope ($p = 0.0478$).

- Days is significant predictor of RFU
- strong upward trend
- R^2 value is 0.866
- slope is statistically significant ($p = 0.048$)

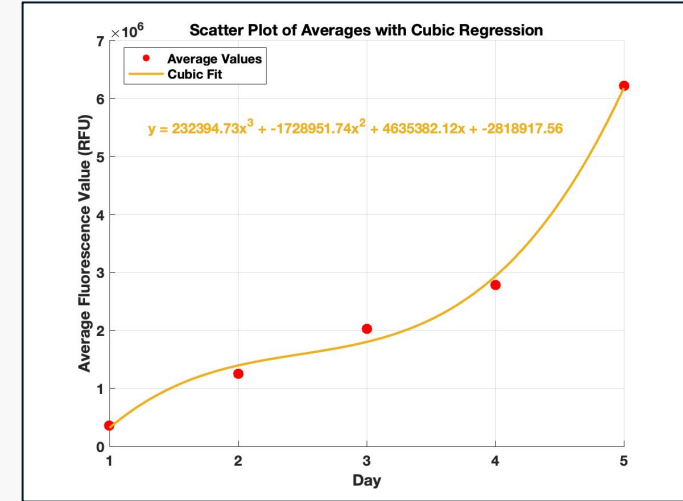


Figure 4: Cubic polynomial regression of average fluorescence values across five days. While the model fits the data closely ($R^2 = 0.995$), the individual coefficients are not statistically significant, likely due to the small sample size.

- capture any non-linear trends
- Cells stabilizing in the middle
- R^2 is very high (0.995) $\Rightarrow$ excellent fit
- not significant, suggesting overfitting or insufficient data
- Best describes curvature, but less significantly reliable for small size

Discussion

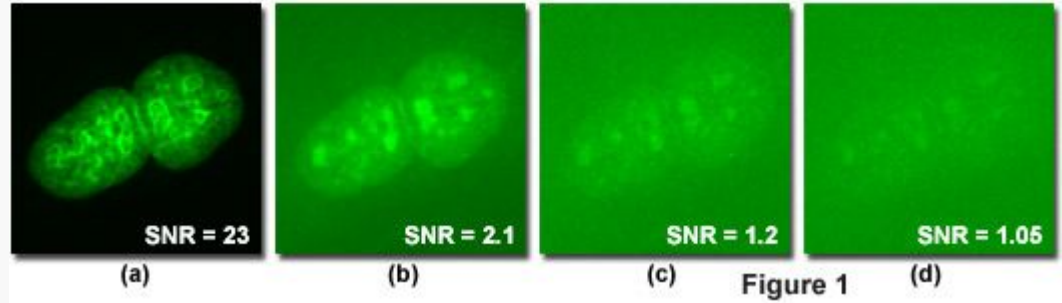
Limitation

- Background noise
- Autofluorescence

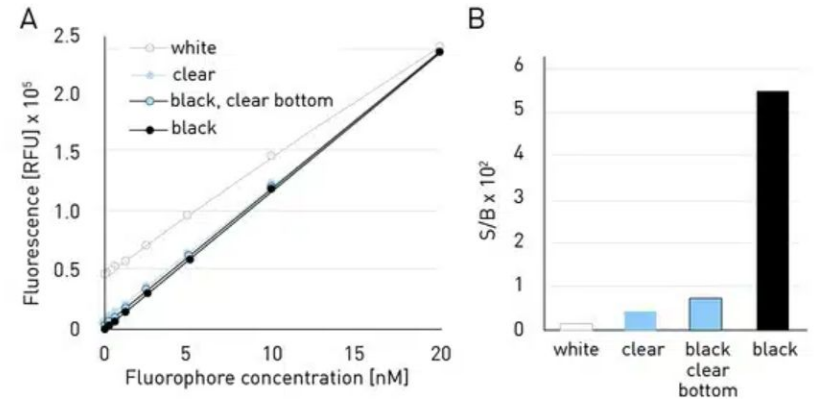
Novel Strategy:

- Dual-wavelength
- Type of plate
- Updated software

Signal-to-Noise Ratios in Fluorescence Microscopy



Molecular Expressions 2025

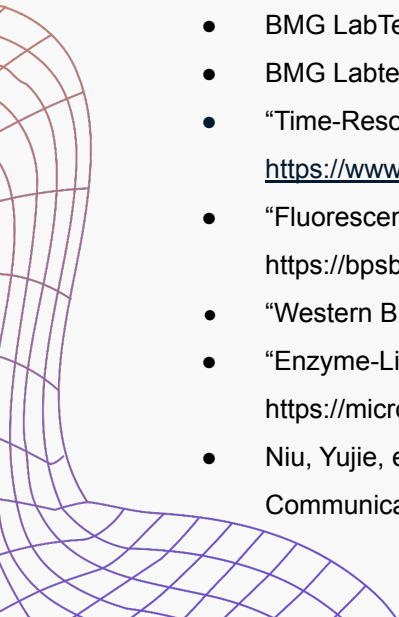


Bite Size Bio 2025



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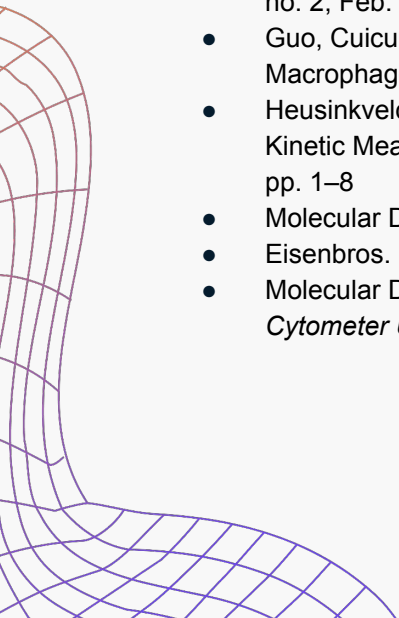
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